

B K Birla Institute of Engineering & Technology, Pilani

45 Days Summer Training Program – Data Science with AIML

Topic: Basic Python | Assignment: 1

Q1. Difference Between Data Types

In Python, data types tell us what kind of value a variable holds. Here are the four basic ones:

a) Integer (int)

Integers are whole numbers – no decimal, no fraction. Can be positive or negative.

```
age = 20
year = 2024
print(type(age)) # Output: <class 'int'>
```

b) Float (float)

Floats are numbers with a decimal point, used when precision matters.

```
price = 99.99
pi = 3.14159
print(type(price)) # Output: <class 'float'>
```

c) String (str)

Strings are sequences of characters — basically any text. Written inside quotes.

```
name = "Gargi"
city = "Jaipur"
print(type(name)) # Output: <class 'str'>
```

d) Boolean (bool)

Booleans can only be True or False. Very useful in conditions and comparisons.

```
is_student = True
has_passed = False
print(type(is_student)) # Output: <class 'bool'>
```

Q2. Creating Variables – name, age, city

```
name = "Gargi"
age = 20
city = "Jaipur"

print("Name:", name)
print("Age:", age)
print("City:", city)
```

Output:

```
Name: Gargi
Age: 20
City: Jaipur
```

Q3. Name Input – Uppercase and Character Count

```
name = input("Enter your name: ")
print("Uppercase:", name.upper())
print("Total characters:", len(name))
```

Sample Output:

```
Enter your name: gargi
```

Uppercase: GARGI
Total characters: 5

Q4. Five Commonly Used String Methods

1. upper()

Converts the entire string to uppercase.

```
s = "hello world"
print(s.upper())    # Output: HELLO WORLD
```

2. lower()

Converts the entire string to lowercase.

```
s = "HELLO"
print(s.lower())    # Output: hello
```

3. strip()

Removes extra spaces from both ends of a string.

```
s = "  hello  "
print(s.strip())    # Output: hello
```

4. replace()

Replaces one part of a string with another.

```
s = "I love cats"
print(s.replace("cats", "dogs"))    # Output: I love dogs
```

5. split()

Splits a string into a list based on a separator.

```
s = "apple,mango,banana"
print(s.split(","))    # Output: ['apple', 'mango', 'banana']
```

Q5. List of Five Fruits

```
fruits = ["apple", "mango", "banana", "grapes", "orange"]

print("Complete list:", fruits)
print("First element:", fruits[0])
print("Last element:", fruits[-1])
print("Total items:", len(fruits))
```

Output:

```
Complete list: ['apple', 'mango', 'banana', 'grapes', 'orange']
First element: apple
Last element: orange
Total items: 5
```

Q6. List Operations – Add, Remove, Print

```
numbers = [10, 20, 30, 40, 50]

numbers.append(60)    # Add 60
numbers.remove(20)    # Remove 20

print("Updated list:", numbers)
```

Output:

```
Updated list: [10, 30, 40, 50, 60]
```

Q7. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is a branch of computer science that focuses on building machines and systems that can perform tasks which normally require human intelligence — things like understanding language, recognizing images, making decisions, and learning from experience.

Why is AI important?

AI is changing the world in a big way. It helps us save time, reduce human error, and solve complex problems faster. In healthcare, it can detect diseases early. In businesses, it automates repetitive tasks. In daily life, it makes things more convenient.

Four Real-Life Applications of AI:

- 1. Healthcare:** AI helps doctors detect cancer and other diseases from medical scans with high accuracy.
- 2. Self-Driving Cars:** Companies like Tesla use AI to analyze road conditions and drive vehicles automatically.
- 3. Recommendation Systems:** Platforms like YouTube and Netflix use AI to suggest videos/movies you'll likely enjoy.
- 4. Virtual Assistants:** Siri, Alexa, and Google Assistant use AI to understand voice commands and respond smartly.

Q8. Which of These Are Examples of AI?

ChatGPT → YES – AI

ChatGPT is a large language model that learns from huge amounts of text data. It understands context, generates human-like responses, and improves through training. This is a classic example of Generative AI.

Google Maps Route Prediction → YES – AI

Google Maps doesn't just follow fixed rules — it learns from real-time traffic data, predicts congestion, and suggests the fastest route. This kind of adaptive decision-making is AI.

Calculator → NO – Not AI

A calculator simply follows fixed mathematical rules. It doesn't learn, adapt, or make decisions on its own. It will always give the same output for the same input — no intelligence involved.

Netflix Recommendations → YES – AI

Netflix uses machine learning algorithms to study your watch history and preferences, then recommends shows and movies you're likely to enjoy. It gets smarter over time — that's AI.

Voice Assistants (Alexa/Siri) → YES – AI

Alexa and Siri use Natural Language Processing (NLP) and machine learning to understand your voice, interpret your intent, and give relevant answers. They also improve with use — definitely AI.